## (a) Global hierarchical pipeline of RAL-MoE-ROM

Parameter $\mu$

**E2 router** (parameter-only)

Preference $\pi$ $[\pi_S, \pi_H, \pi_P]$

**Top-1 or Top-2**

S-H and H-P pairs only

**Top-1: S, H, or P** Autonomous rollout

Physical reconstruction

**Predicted fields**

Fusion after reconstruction in common physical space

Top-2

**Regime-local specialists**

**Steady** $S$ | **Hopf** $H$ | **Periodic** $P$

Independent local reduced charts

**Specialist A** Independent rollout

Physical reconstruction

**Specialist B** Independent rollout

Physical reconstruction

**Physical space T2-C fusion**

**Physical history** $H_n$

Initialize selected native chart(s)

From physical history

Chart-independent descriptor $d_n$

$\mu$, pair preference $\pi^-$

**T2-C weight** $\alpha$ correction $\Delta_\psi(\mu, d_n)$

**Predicted fields**

## (b) Zoom-in: one regime-local specialist

Local reduced chart

**Physical history** $H_n$

**Local encoder**

Local state / history

Features $\xi_n$

**Sparse MoE velocity correction**

**Group router**

**Shared expert**

**Top-2 experts**

Weighted sum

Nonlinear + affine + low-rank bilinear
Shared and routed experts
within the selected group

**Projected velocity backbone** $F_r^G$

Updated local prediction $(a_{n+1}, b_{n+1})$

Velocity derivative

**Velocity time integration** (RK)

**Velocity** $a_{n+1}$

Algebraic pressure reconstruction (local features)

**Pressure** $b_{n+1}$